Toheeb Babatunde Ibrahim

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PROFESSIONAL SUMMARY

Doctoral researcher in Pure and Applied Mathematics at Bowling Green State University, with interdisciplinary training in mathematical engineering and nanotechnology. Develops physics-based and control-oriented models for industrial glass production, applying optimization, COMSOL, and MATLAB to improve process efficiency and product performance. Experienced in modeling industrial, financial, and biological systems and collaborating across applied mathematics, materials science, and process engineering to translate complex challenges into precise, research-driven solutions.

TECHNICAL SKILLS

Programming & Data: Python, MATLAB, R, SQL

Simulation & CAD: ANSYS, Onshape, LibreCAD, LAMMPS/VMD

Methods: Dynamic systems, ODE/PDE modeling, numerical analysis, convex optimization, stochastic modeling, PID/FOPID, LQR, state estimation, model reduction, physics-informed neural networks

Research Tools: LaTeX, GeoGebra, Desmos, technical reporting, scientific reporting

EDUCATION

Bowling Green State University — **Aug. 2024 – Present**
Doctor of Philosophy in Applied Mathematics — *Bowling Green, Ohio, US*

Gdańsk University of Technology — **Oct. 2023 – Jan. 2025**
Master of Science in Nanotechnology (Dóbry Plus) — *Gdańsk, Poland*

University of L'Aquila — **Sep. 2022 – Sep. 2023**
Master of Science in Mathematical Engineering (108/110) — *L'Aquila, Italy*

African Institute for Mathematical Sciences — **Oct. 2018 – Jun. 2019**
Master of Mathematical Sciences (Very Good Pass) — *Limbe, Cameroon*

University of Ilorin — **Sep. 2013 – Oct. 2017**
Bachelor of Science in Mathematics (First Class Honors) — *Ilorin, Nigeria*

WORK EXPERIENCE

College of Engineering, Bowling Green State University — **August 2025 – Present**
Research Assistant — *Bowling Green, Ohio, US*

- Develop a physics-based control system model for an industrial glass melting furnace
- Analysed control architecture including PID, LQR and observer-based control.
- Implemented advanced controller tuning technique, including PSO-tuned PID/FOPID.
- Study the mathematical theory in the model reduction of CFD models to simpler mathematical/control model
- Organize and cordinating meeting schedules with industrial patners.

Department of Mathematics & Statistics, Bowling Green State University — **August 2024 – May 2025**
Graduate Teaching Associate — *Bowling Green, Ohio, US*

- Taught first-year students College algebra in the Math Emporium.
- Taught Calculus and Analytical Geometry to Biology and Pre-Med students
- Cordinate the course and implemented the course curriculum to the student.
- Implemented a computer assisted teaching e.g. Geogebra, Desmos etc for visualization of mathematical concepts.

African Institute for Mathematical Science — **Nov. 2019 – Dec 2019**
Teaching Assistant — *Limbe, Cameroon*

- Assisted in teaching Mathematical Modeling of Dynamical Systems by providing tutorial for the student of the course
- Guided the student in the use of Berkeley Madonna for the simulation of their dynamical systems model.
- Guided the student on the interpretation of model and translating their result into policy.

Adekunle Ajasin University **Dec. 2017 – Oct. 2018**
Teaching Assistant *Akungba-Akoko, Nigeria*

- Tutored First-year students in the departments of mathematics on foundational mathematics.
- Assisted final year students with MATLAB simulation

RESEARCH PROJECTS

Hybrid Neural ODE-Based Modeling and Observer-Based Regulation of Industrial Glass Melters2026

- Developed a first-principles model of glass-melting process dynamics and a Luenberger observer for estimating states required in LQR design
- Worked with researchers in applied mathematics, chemical engineering, and industrial glass manufacturing

Stochastic Mortality and Social Security Progressivity **2026**

- Investigated actuarial models for Social Security benefits and analyzed how economic-dependent mortality can affect benefit calculations and progressivity

Central Force Decomposition of the 2NN-MEAM Potential **2024**

- Conducted master's research at Gdańsk University of Technology on the mathematical decomposition of the second-nearest-neighbor modified embedded-atom-method potential

COVID-19 Epidemiological Modeling Collaboratons **2021**

- Contributed to mathematical modeling studies of repeated viral exposure and vaccination campaigns, resulting in two peer-reviewed PLOS ONE publications.

PUBLICATION

Kristina B.H., Girma M.Z, **Toheeb B.I.**, Aliou B., Henri T.O., Vicent A., Gideon A. N., Miranda T. E, Kristan A.S. *Is increased mortality by multiple exposures to COVID-19 an overseen factor when aiming for herd immunity?*. PLOS One (2021). Link

Nessma A.Y., Henri T.O., Yvan Jordan N.M., Sandy K.N., Loyce K., George K., **Toheeb B.I.**, Patience B.I., Sulyman I., Looli A.N., Kristina B.H., Miranda T.E. Kristan A.S. *The impact of COVID-19 vaccination campaigns accounting for antibody-dependent enhancement*. PLOS One (2021).Link

SCHOLARSHIPS AND AWARDS

J. Robert & G. Overman Scholarship Award, Bowling Green State University 2025
Erasmus+ Scholarship 2022 – 2024
DAAD Research Scholarship 2019 – 2021
Petroleum Technology Development Fund Scholarship 2016
Gold Medal, National Mathematics Competition for University Students 2016

ADDITIONAL INFORMATION

Professional Development: AI-Ready Ohio (2026); Great Lakes AI Week Conference (2025)
Languages: English (full professional), Yoruba (native), Italian (A2), French (A2), Polish (A1)
Professional Membership: American Mathematical Society

REFERENCES

Available upon request